Hints: Inequalities – 1. The Cauchy–Bunyakovsky inequality as a technique

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Discrete form: $P(\lambda) = \sum_i (a_i - \lambda b_i)^2 \geqslant 0$ for all real λ is a quadratic in λ (if not all $b_i = 0$), so its discriminant $4\big(\sum a_i b_i\big)^2 - 4\sum a_i^2 \sum b_i^2$ is $\leqslant 0$. Equality means P has a real root λ, i.e. $a_i = \lambda b_i$ for all i.

Hint 2. The integral form and the inner product form are word for word the same with $P(\lambda) = \int (f - \lambda g)^2$ and $P(\lambda) = \langle f - \lambda g, f - \lambda g\rangle$.

Hint 3. Engel form: apply the discrete inequality to the vectors $(a_i/\sqrt{b_i})$ and $(\sqrt{b_i})$: $\big(\sum a_i\big)^2 = \big(\sum \frac{a_i}{\sqrt{b_i}}\sqrt{b_i}\big)^2 \leqslant \sum \frac{a_i^2}{b_i} \sum b_i$.

Problem 2

Hint 1. Use Fact 1: the data are $\int f \cdot 1$ and $\int f \cdot x$, so the extremal function should be linear. Find $f_0(x) = ax + b$ with $\int_0^1 f_0 = \int_0^1 x f_0 = 1$ and apply Cauchy–Bunyakovsky to f and f_0.

Hint 2. $\frac{a}{2} + b = 1$, $\frac{a}{3} + \frac{b}{2} = 1$ give $f_0 = 6x - 2$. Then $\int_0^1 f f_0 = 6\int_0^1 xf - 2\int_0^1 f = 4$ and $\int_0^1 f_0^2 = 12 - 12 + 4 = 4$, so $16 \leqslant 4\int_0^1 f^2$, i.e. $\int_0^1 f^2 \geqslant 4$, with equality for $f(x) = 6x - 2$. (Equivalently: $0 \leqslant \int_0^1 (f - f_0)^2 = \int_0^1 f^2 - 8 + 4$.)

Problem 3

Hint 1. By linearity the conditions say $\int_0^1 q(x) f(x)\,dx = q(1)$ for every polynomial q of degree $< n$. So compare f with the polynomials of degree $< n$ via an orthonormal basis of that space.

Hint 2. The shifted Legendre polynomials $e_k(x) = \sqrt{2k+1}\,P_k(2x-1)$, $k = 0, \dots, n-1$, are orthonormal in $L^2[0,1]$ (since $\int_{-1}^1 P_k P_l = \frac{2\delta_{kl}}{2k+1}$) and $P_k(1) = 1$. Hence $\langle f, e_k\rangle = e_k(1) = \sqrt{2k+1}$.

Hint 3. Put $p = \sum_{k<n} e_k(1) e_k$. Then $\int_0^1 fp = \sum_{k<n} e_k(1)^2 = \sum_{k<n}(2k+1) = n^2$ and $\int_0^1 p^2 = n^2$ as well, so Cauchy–Bunyakovsky gives $n^4 \leqslant n^2 \int_0^1 f^2$, i.e. $\int_0^1 f^2 \geqslant n^2$ (this is Bessel's inequality), with equality for $f = p$. For $n = 2$: $p = 1 + 3(2x-1) = 6x - 2$, and the conditions $\int f = \int xf = 1$ are those of Problem 2.

Problem 4

Hint 1. Substitute $a_i = \frac{1}{1+x_i}$: then $\sum a_i = 1$ and $x_i = \frac{1-a_i}{a_i} = \frac{\sum_{j\neq i} a_j}{a_i}$.

Hint 2. Lower bound the left side: $\sqrt{\sum_{j\neq i} a_j} \geqslant \frac{1}{\sqrt{n-1}} \sum_{j\neq i} \sqrt{a_j}$, so

$$\sum_i \sqrt{x_i} \geqslant \frac{1}{\sqrt{n-1}} \sum_j \sqrt{a_j} \sum_{i\neq j} \frac{1}{\sqrt{a_i}}$$

(swap the order of summation).

Hint 3. By the Engel form, $\sum_{i\neq j}\frac{1}{\sqrt{a_i}} \geqslant \frac{(n-1)^2}{\sum_{i\neq j}\sqrt{a_i}}$, and by Cauchy–Bunyakovsky again $\sum_{i\neq j}\sqrt{a_i} \leqslant \sqrt{(n-1)\sum_{i\neq j}a_i}$. Put these together: $\frac{\sqrt{a_j}}{\sqrt{\sum_{i\neq j}a_i}} = \frac{1}{\sqrt{x_j}}$, and the constants combine to $(n-1)$. Equality holds iff all $x_i = n-1$ (for $n=2$ the condition means $x_1x_2=1$ and the inequality is an identity).

Problem 5

Hint 1. Use $abc=1$ to rewrite each term as a square over something: $\frac{1}{a^3(b+c)} = \frac{(bc)^2}{a(b+c)}$.

Hint 2. Engel form: $\sum\frac{(bc)^2}{ab+ac} \geqslant \frac{(ab+bc+ca)^2}{2(ab+bc+ca)} = \frac{ab+bc+ca}{2}$. It remains to prove $x+y+z\geqslant 3$ for $x=ab$, $y=bc$, $z=ca$, $xyz=1$, without AM-GM.

Hint 3. Since $xyz=1$, $xy+yz+zx=\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$, so Cauchy–Bunyakovsky gives $(x+y+z)(xy+yz+zx)\geqslant 9$. Also $(x+y+z)^2\geqslant 3(xy+yz+zx)$ (from $x^2+y^2+z^2\geqslant xy+yz+zx$, again Cauchy–Bunyakovsky). Multiply: $(x+y+z)^3\geqslant 27$. Equality holds iff $a=b=c=1$.